SCHEDULA • HOSPITAL MANAGEMENT SYSTEM

Schedule Project Module • Complete 5-Entity Relational Architecture

1. User Module | 2. Patient Module | 3. Doctor Module | 4. Availability Module | 5. Appointment Module

[M1] USERS

PK	**id**	UUID
REQ	email (UNIQUE)	VARCHAR
OPT	phone (UNIQUE)	VARCHAR
OPT	password_hash	VARCHAR
REQ	role [DOC|PAT|ADM]	VARCHAR
REQ	auth_provider	VARCHAR
REQ	is_verified	BOOLEAN
SYS	created_at	TIMESTAMP
SYS	updated_at	TIMESTAMP

identifies (1:1)

[M2] PATIENTS

PK	**id**	UUID
FK	user_id (1:1)	UUID
REQ	first_name	VARCHAR
REQ	last_name	VARCHAR
REQ	date_of_birth	DATE
REQ	gender [M|F|O]	VARCHAR
OPT	blood_group	VARCHAR
OPT	emergency_contact	VARCHAR
OPT	address	TEXT
OPT	medical_history	TEXT
SYS	created_at	TIMESTAMP

books (1:N)

[M5] APPOINTMENTS

PK	**id**	UUID
FK	patient_id	UUID
FK	doctor_id	UUID
REQ	appointment_date	DATE
REQ	start_time	TIME
REQ	end_time	TIME
REQ	status [CONF|CANC|...]	VARCHAR
OPT	reason (Symptoms)	TEXT
OPT	cancellation_reason	TEXT
OPT	reschedule_reason	TEXT
SYS	created_at	TIMESTAMP
SYS	updated_at	TIMESTAMP

identifies (1:1)

[M3] DOCTORS

PK	**id**	UUID
FK	user_id (1:1)	UUID
REQ	first_name	VARCHAR
REQ	last_name	VARCHAR
REQ	specialization	VARCHAR
REQ	license_number	VARCHAR
REQ	experience_years	INT
REQ	consultation_fee	DECIMAL
OPT	bio	TEXT
REQ	rating_average	DECIMAL
REQ	is_approved	BOOLEAN

configures (1:N)

[M4] DOCTOR_AVAILABILITY

attends (1:N)

PK	**id**	UUID
FK	doctor_id	UUID
REQ	day_of_week	VARCHAR
REQ	start_time	TIME
REQ	end_time	TIME
REQ	slot_duration_minutes	INT
REQ	max_capacity (Elastic)	INT
REQ	is_active	BOOLEAN
SYS	created_at	TIMESTAMP

SCHEDULA • DATA DICTIONARY & SPECIFICATIONS (5 ENTITIES)

Schedule Project Module • Complete Attribute Specifications, Constraints & Relationships

Module	Entity	Key Columns	Data Type	Constraints & References	Description
1. User	users	id, email, phone, password_hash, role, auth_provider, is_verified	UUID, VARCHAR(255), VARCHAR(20), BOOLEAN, TIMESTAMPTZ	PK, UNIQUE(email, phone) role: DOCTOR | PATIENT | ADMIN auth_provider: LOCAL | GOOGLE	Central authentication identity; manages passwords, OAuth & OTP verification
2. Patient	patients	id, user_id, first_name, last_name, date_of_birth, gender, blood_group, address, medical_history	UUID, VARCHAR(100), DATE, VARCHAR(20), TEXT, TIMESTAMPTZ	PK, FK -> users(id) UNIQUE (1:1) ON DELETE CASCADE	Patient demographic identity, emergency contacts, and baseline clinical history
3. Doctor	doctors	id, user_id, first_name, last_name, specialization, license_number, fee, bio, rating_average, is_approved	UUID, VARCHAR(100), VARCHAR(50), DECIMAL(10,2), INT, TEXT, BOOLEAN	PK, FK -> users(id) UNIQUE (1:1) UNIQUE(license_number)	Doctor professional credentials, verified medical license, consultation fee, ratings & admin approval
4. Availability	doctor_availability	id, doctor_id, day_of_week, start_time, end_time, slot_duration_minutes, max_capacity, is_active	UUID, VARCHAR(20), TIME, INT, BOOLEAN, TIMESTAMPTZ	PK, FK -> doctors(id) ON DELETE CASCADE max_capacity >= 1 (Elastic)	Doctor weekly consultation windows, slot durations, and elastic capacity rules for walk-ins
5. Appointment	appointments	id, patient_id, doctor_id, appointment_date, start_time, end_time, status, reason, cancellation_reason, reschedule_reason	UUID, DATE, TIME, VARCHAR(30), TEXT, TIMESTAMPTZ	PK, FK -> patients(id) FK -> doctors(id) status: SCHED | CONF | CANC | RESCHED	Core booking transaction capturing scheduled slot, patient symptoms, cancellation and reschedule notes